Time-Series Forecasting of Household Appliance Energy Use: A Comparative Model Case Study

Github Link: https://github.com/MoazTahir/Timeseries-Datascience **Student ID**: 24178436

1. Introduction and Overview

This report aims on dissecting and understanding household appliance energy usage prediction using the UCI Appliances Energy Prediction dataset (Candanedo, Feldheim and Deramaix, 2017). The dataset consists of hourly records of appliance usage from a low energy utilization house located in Stamburges, Belgium. Apart from appliances, indoor sensor and outdoor weather covariates are also present as additional datapoints. Using the data provided, the ideal prediction scenario is a 24 hour ahead forecasting of all the household appliances measured using Watthour utilization per hour (Wh) which was done with evaluation with a 14-day rolling window across 6 model families and five statistical benchmarks. The benchmarks were namely mean, naïve, daily/weekly seasonal naïve, drift. As for the models we will look into SARIMAX family and a zero shot foundation time series model Chronos.

All the methods tested contributed to a different hypothesis. The main goal of each benchmark was to establish if a series is dominated by a recurring routine or by a short-term momentum. Seasonality (on a daily level) and autocorrelation are tested by SARIMAX in both exogenous and non exogeneous weather, as to see if that explicit model can improve on that routine or not. To test how things work with in terms of signal extraction from hand engineered features, XGBoost has been used. Lastly the Chronos model is there to experiment and see whether a model that is pretrained on some other time series data can match or beat models that are trained for this specific task. Every single section mentioned below explains and argues on how the results came to be using analytical figures that were generated alongside the models' outputs.

2. Dataset and Preparation

19,735 rows of 10-minute intervals from 11 Jan to 27 May 2016 are present that belong to a m-bus energy meter of the target. This also includes nine indoor temperature and humidity sensor pairs and 6 outdoor weather variables (not one onsite sensor, these were in fact, sourced by nearest official weather station to the location mentioned above). This would further prove to showcase the weaker correlation to the target variable as seen in section 3.

The rv1/rv2 columns are synthetic random columns. These were dropped. With these columns, another three were dropped that had deterministic functions of timestamps, as these were already in the index column. Upon further exploration, no missing values were found so no dropping or imputation was needed. The series was made up with 10 minutes to hourly samples, made via aggregation. If we were to implement a SARIMAX fit on that, a period-24 seasonal term costs 10-30 seconds for hourly data. However, a 10-minute resolution would need to be a period-144 and would have 147 combinations. On one hand, the previous one took about 13-23 minutes on AIC search. On the other hand, hourly data would've plausibly taken hours rather than the time mentioned above. The hourly series after processing contains 3290 rows and the mean value is 97.78 Wh, std dev value is 81.21, median value is 63.33 and the max value is 608.33 Wh. It can be noted that the mean is 50% above the median dataset that strongly suggests a right skewed nature of the dataset. That means most of the hours are quiet with some rare high usage hours. This asymmetry is found throughout the dataset and even ranges from SARIMAX residual tail (more on that in section 5) up till foundation models quantile average (more on that in section 7).

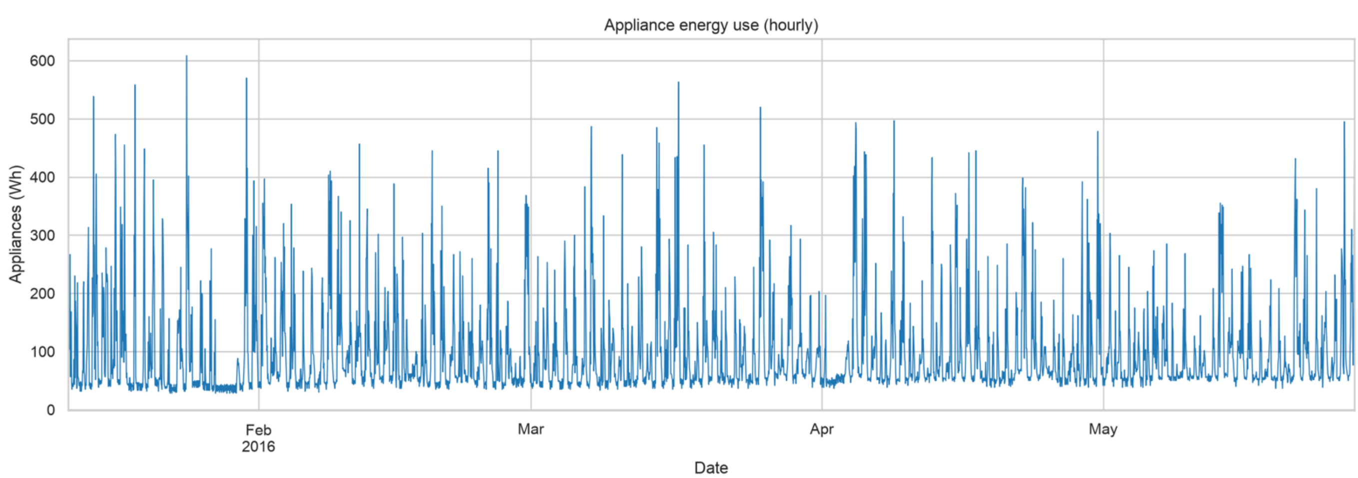


Figure 1. Full 4.5-month hourly series. An approximate 10-day low-occupancy period in late January (right before the Feb 2016 x-axis label) sits inside the training data, which is a caveat in data representativeness that would further be discussed in Section 10.

From the total data, the remaining 14 days (roughly 336 hours) aren't included in the above data as that would be used in the test set. The figure 2 below shows that this window isn't uniformly difficult. Looking at the below data, the first week that spans 13-20 may has very modest and regular peaks whilst the week right after that contains 4 alarmingly large events of power usage. On 21 may there was 430WH usage, on May 23-24 that was 340-380. The largest dataset value comes from the power usage on 26-27 may that was in 500 Wh range. This evidence is properly visual that the test is harder in the second half as there were shared spike errors across every model in that area (more on that later in section 8)

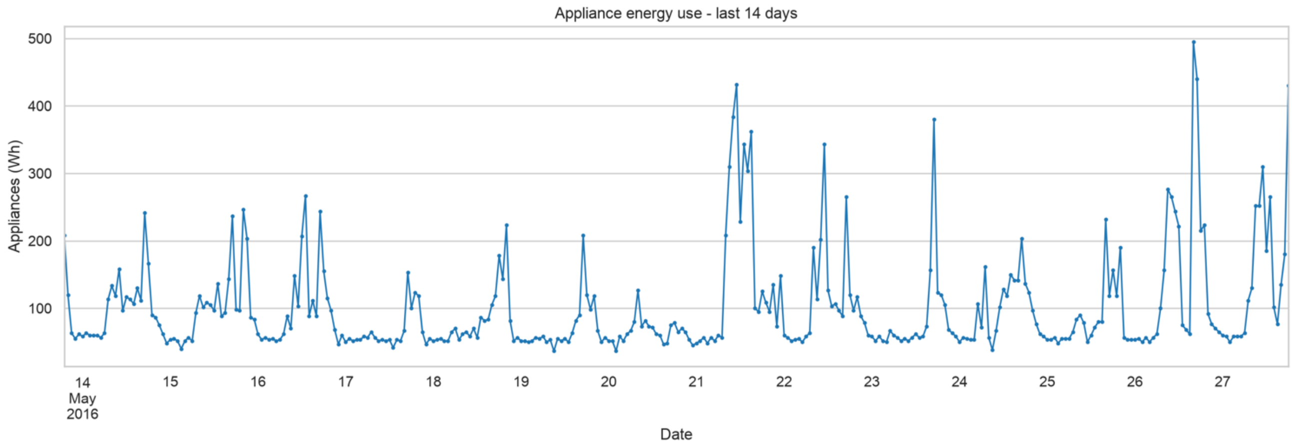


Figure 2. Test window from 13-27 May with hourly points. The second week contains three-four distinct large spike events.

3. Exploratory Analysis and Stationarity

The figure 3 below illustrates the daily cycles (left) and weekly cycles (right) in separated panels. The hour of day panel has a strong pattern where median is flat around the 45-55 Wh range at nighttime. However, there is a trend change as we move to morning time as there is a sharp rise from hour 7 that peaks around the 18 (6:00pm) where the median lies in the 165 range. The most extreme outliers and widest boxes are present in the 8-20 band range. As for the day of week panel, it is comparatively flat with median values in 55-70 range. It is to note that Friday and Saturday have a bit wider boxes and higher upper quartiles than the Tues-Thursday band range. This daily cycle has a dominant seasonal effect as the weekly effect is real but more subtle. This would be looked more into when interpreting benchmark results in section 4.

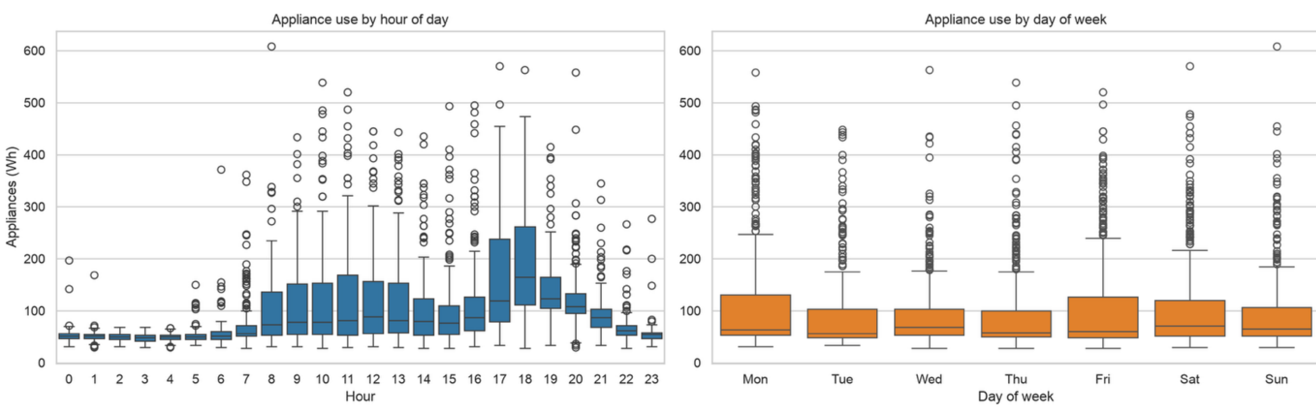


Figure 3. Hour-of-day(left) & day-of-week (right) usage profiles (used in training period).

Additionally, ADF and KPSS tests were run on different series as well. Initially on level series, and then first, seasonal and combined differences (as illustrated in the table below). Both tests agree that the series is in fact stationary in nature even at the raw level. The ADF null comes out to be a unit root while KPSS null is stationary. Both these appear to contradict the usual strong seasonal cycles that have been illustrated in the figures above, but they do not. These ADF/KPSS test check narrowly for a stochastic trend. This does not effectively tell the outcome for a deterministic repeating cycle. That means that a series can be stationary in the specific sense whilst still having some seasoning term in the forecasting model. This is exactly why the SARIMAX would still require a seasonal component. Regardless of the AIC optimal order having d=0 as the parameter.

Series	ADF stat	ADF p	KPSS stat	KPSS p
Level	-8.95	8.8e-15	0.038	≥0.1
First diff	-16.95	9.4e-30	0.078	≥0.1
Seasonal diff (lag 24)	-13.59	2.0e-25	0.009	≥0.1
First + seasonal diff	-18.15	2.5e-30	0.081	≥0.1

Table 1. Stationarity tests (n ≈ 3,230-3,260 per row).

The 4 plots below show the ACF/PACF mentioned earlier. The changes are in pro vs pre seasonal differencing. On the ACF side, the ACF decays from 0.58 at the initial lag till the next lag timestamp at 12 (0.17 which is roughly a half a day of phase offset). The secondary peak of 0.30 is reached at the 3rd lag of 24 which is the daily seasonality signature (signal of new day). The decay keeps on repeating from that point with decay amplitude at the 48 and 72 timestamps. Talking about PACF, it cuts off sharply after the second lag. However, it retains a small seasonal bump near the end of day lag of 21-24. After just one seasonal difference, the short lag structure from before does survive but remains largely unchanged. This is a key information to note as this signals a non-seasonal AR of order p6. On the other hand, the positive lag-24 peak becomes a negative spike of roughly -0,44. That is an expected aftermath of trying to difference series with positive lag 24 autocorrelation and that the seasonal MA term is designed to absorb.

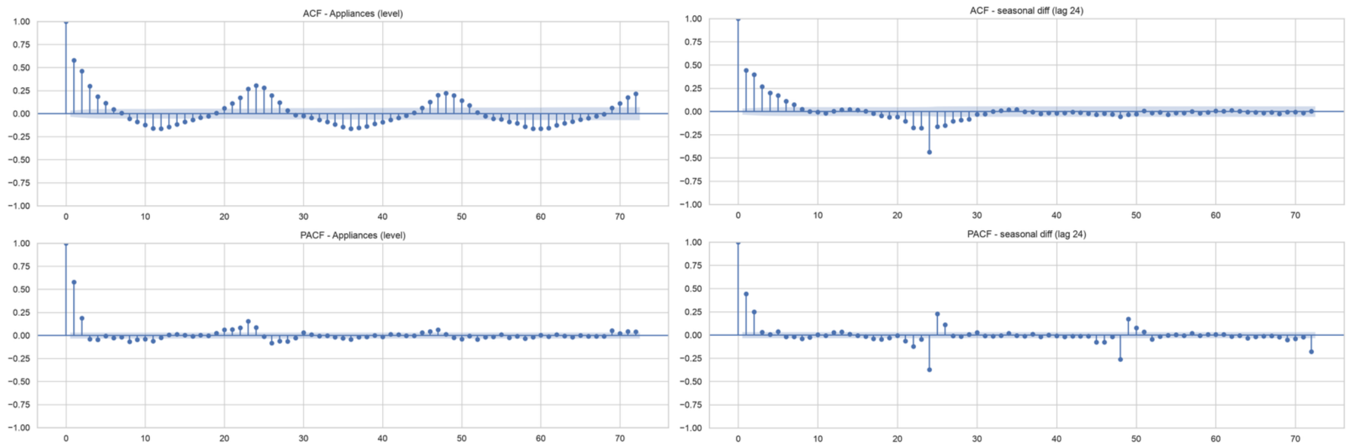


Figure 4. ACF/PACF of the level series and after seasonal (lag-24) differencing.

The heatmap figure 5 shows a very weak target covariate correlation throughout all features (sub 0.3) but a very strong correlation on T_out and Tdewpoint (0.8 range) and on the three indoor sensors (T1, T2, T3) with values in range of 0.9-0.95. This multi collinearity without a relevant pattern not only shows a strong covariate to covariate correlation, but also shows how weakly informative it is about the target that makes it the root cause evidence for the two later findings we would look into in later sections (SARIMAX exogenous variant underperformance & sensor/weather feature making the XGBoost worser, Section 5+6)

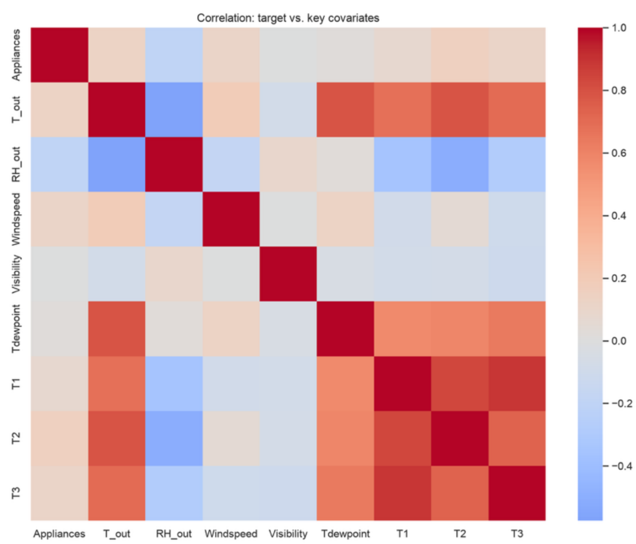


Figure 5. Correlation matrix showcasing target and covariates.

4. Benchmark Models

Below are the evaluation of 5 model baselines using the same rolling origin as all the others. Those being mean, naive (last observed value), daily seasonal-naive (lag 24), weekly seasonal-naive (lag 168) and drift. (linear extrapolation).

Model	MAE	RMSE	MASE	Bias
Weekly seasonal-naive	43.46	81.41	0.813	-13.16
Daily seasonal-naive	48.31	85.56	0.904	+1.75
Mean	50.26	74.94	0.941	-3.29
Naive	85.55	110.39	1.601	+50.98
Drift	85.80	110.68	1.606	+51.37

Table 2. Benchmark results, containing all 14 rolling-origin test days.

Naïve and drift values are twice as bad as others by using MASE. Both values also carry a large positive bias of 51. That refers to one thing, both extrapolate from wherever the training window is ending that usually refers to the moment where the power usage is in the moderate to high usage tier. These levels are held out into hours that are usually much quitter overnight. When comparing seasonal naïve, the weekly beat daily ones at 0.813 vs 0.904. This led to one important result, the repeating value from a week ago matches both time of day and day of week. This confirms that weekday/weekend effect that was apparent in the figure 3. This also tells us that the dominant structure of the trend exists and follows the day of week and time of day. That also reinforces that short term momentum isn't the contributing factor.

5. SARIMAX

With the seasonal order fixed at 1,1,1,24 (which is derived from STL decomposition and the ACF comparisons of figure 4, rather than a jointly search), a gird search over the p,d,q values at 147 combinations ((p,d,q) ∈ [0,6]×[0,2]×[0,6]) was run. Jointly trying to search even 3 seasonal values would've multiplied the grid by roughly 27 times on top of the already 10-30 seconds cost per fit. Out of 147 candidates, 50 converged. Out of all the orders, selected one was (6, 0, 0) that has an AIC of 32,161. The surface of AIC at d=0 is mostly flat across all combinations (32,160-32,660). There however was a degenerate outlier at p=4 and q=2 with AIC at roughly 68,342. This confirms p=6 and q=0 to be a solid, although not dramatically better than other alternatives.

In the residual diagnosis figure below the Q-Q plot is tracking a normal line that goes through the middle whilst cutting sharply above at its upper tail. This showcases a heavily tailed right skewed signature mirroring the targets own skew as informed about beforehand in section 2. This formally confirms what the later forecasting plots would show. Model is bad on predicting spikes, often under predicting and rarely overpredicting by a comparable margin.

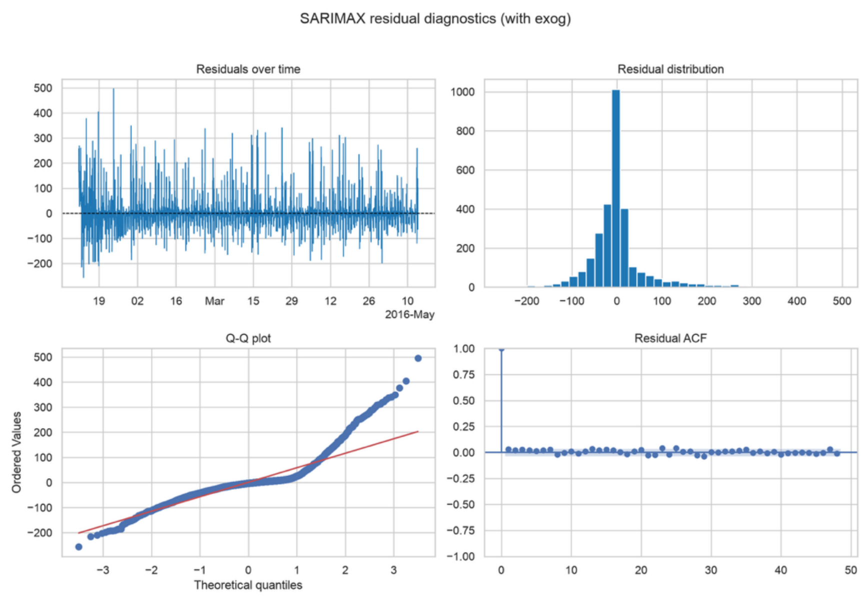
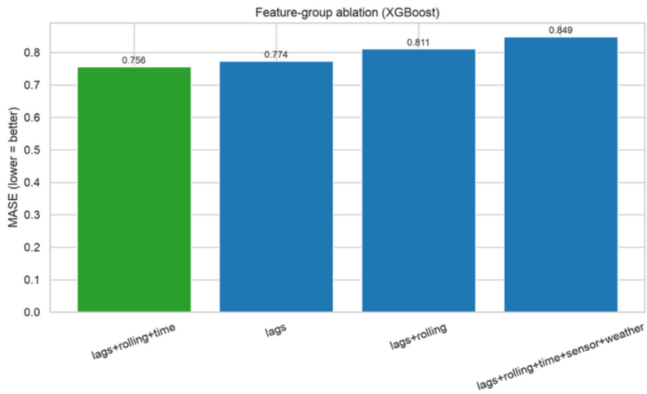


Figure 7. SARIMAX (6,0,0) (1,1,1,24) training-residual diagnostics.

When the exogeneous weather information is added (namely T_out, RH_out, Windspeed, Visibility, Tdewpoint data points), the covariates made SARIMAX slightly worser as the MASE went from 0.676 to 0.687. Most likely it caused the model to further overfit. The weather variables do however correlate weakly with appliances but strongly with each other. That means there are more parameters being added, but they have no real signal associated with them. Given how the figure 5 from above has weak target correlation and strong inter correlation, having those 5 more parameters to re-estimate the 2.9k points once again gives no new signal gets consistent with degradation of noise fitting rather than a real generalizable weather effect.

6. Feature Engineering and the Feature-Based Model

In total, 5 extra features were engineered. Those five were time, lag, mean std, sensor pairs, and weather columns. To have a genuine 24-hour forecast, you need a short lag/rolling feature that falls inside the horizon and hasn't been observed yet at the time of forecast. This is to make sure that the model steps through the horizon recursively by building each hour features from real history with the own earlier predictions. This is also verified via deterministic stub model through a dedicated test. Moving to the real-life testing scores, XGBoost with full features scored MASE of 0.849 (RMSE of 65.63 and bias of +13.83), just ahead of HistGradientBoosting Regressor (0.863). It should be noted that lags only score 0.774 and having the rolling window worsens it to 0.811. However, if you add time of day on tip, it improves past the lag baseline which turns out to be the best of four configs. However, adding sensor and weather features makes the full model worser,



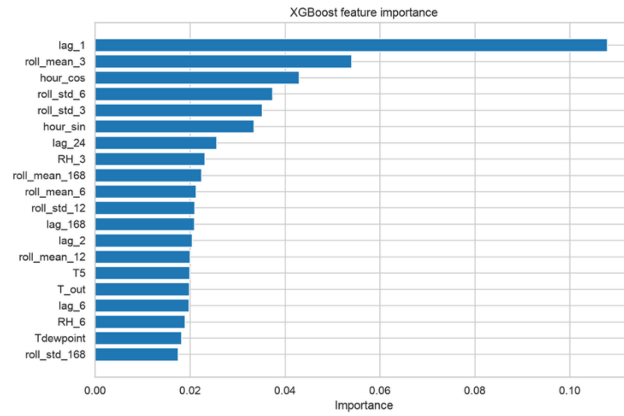


Figure 9 (Left). XGBoost feature-group ablation; the deployed full-feature model is the worst of four configurations.

Figure 8 (Right). Top 20 XGBoost feature importances (full-feature model).

Looking at the top features, lag_1 is the most important one as it is roughly double the roll mean 3, the next contender. All the top 6 are the lag rolling or some time-based features. The sensor or weather column don't show up until the 8th place at the fifth of lag 1s importance. There are two things to focus on, lag and time features carry the real signal and sensor. Weather features are individually marginal in contribution and collectively they dilute the performance. Which simply classifies them as extra features without extra data increase in variance without a proportionate bias reduction.

7. Foundation Model

We have investigated SARIMA, now let's look at the foundational model that has been trained on unrelated another dataset called Chronos-Bolt-tiny (Ansari et al., 2024). This model was used zero shot with zero finetuning on this data. It was provided with most recent 512 hours of data at its inference time. The other options were TimeGPT (paid) and TimesFM (very heavy local setup) as it can run entirely on CPU. However, unlike all the other models being used this has no conditioning to nay covariate value such as time of day.

Chronons got MASE of 0.649 and RMSE of 68.31. This is by far the best MASE out of all the models so far despite having no training done and it being zero shot. The below figure showcases its rhythm tracking on forecasting. Although it does fail on reaching large spikes that are much apparent in the second weeks (median staying around 100-150 while the actual is around 430-500Wh) The 10-90th percentile gives it an edge as those bands cover most peaks even when the point forecast is a hit or miss. This can end up being useful for sizing a battery shortage or some load shedding threshold. This was not possible with the point only models. Coming back once more to the right skewed target, this may turn out to be useful and much better than the SARIMAX's symmetric intervals, although it hasn't been extensively tested here and is a well scoped future addition which is mentioned in section 10.

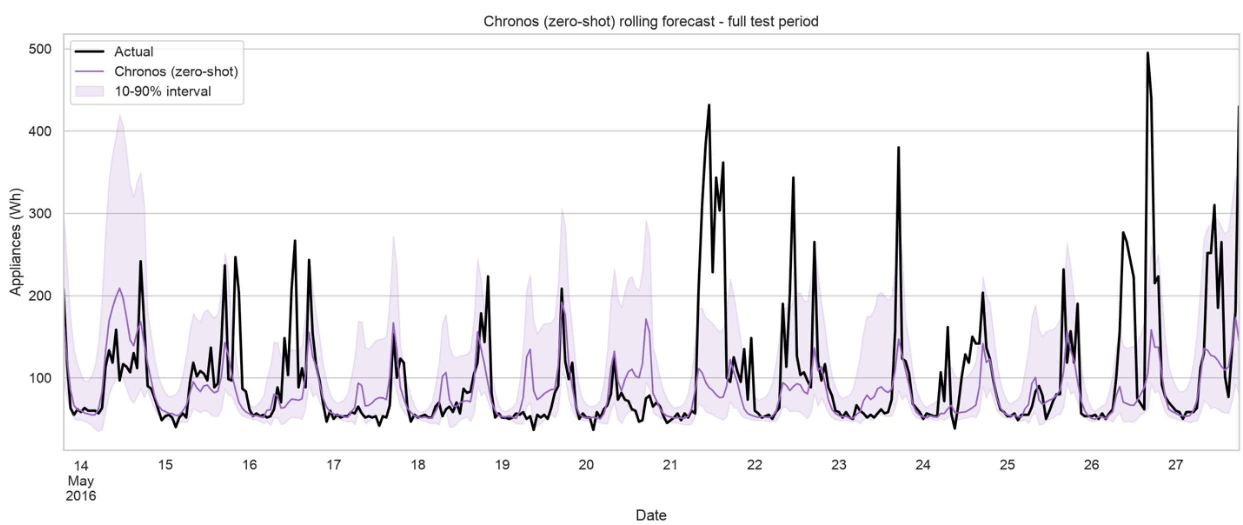


Figure 10. Chronos-Bolt-tiny zero-shot forecast with 10th-90th percentile band,

This model outperforming others without any access to data can be tied back to the section 4s thesis where the series dominant structure is a generically re-occurring routine and a well pretrained model such as Chronos would easily detect that exact pattern and generalize better. Older models were rederiving a pattern that this pretrained model already comes trained with.

8. Full Comparison and Results

Rank	Model	MAE	RMSE	MASE	Bias
1	Chronos	34.67	68.31	0.649	-17.10
2	SARIMAX (univ.)	36.11	64.69	0.676	-7.24
3	SARIMAX (+exog)	36.70	64.79	0.687	-5.02
4	Seas-naive (weekly)	43.46	81.41	0.813	-13.16
5	XGBoost (full feat.)	45.34	65.63	0.849	+13.83
6	HistGB	46.11	67.01	0.863	+16.00
7	Seas-naive (daily)	48.31	85.56	0.904	+1.75
8	Mean	50.26	74.94	0.941	-3.29
9	Naive	85.55	110.39	1.601	+50.98
10	Drift	85.80	110.68	1.606	+51.37

Table 3. All ten models, MASE-sorted, full 14-day rolling-origin backtest.

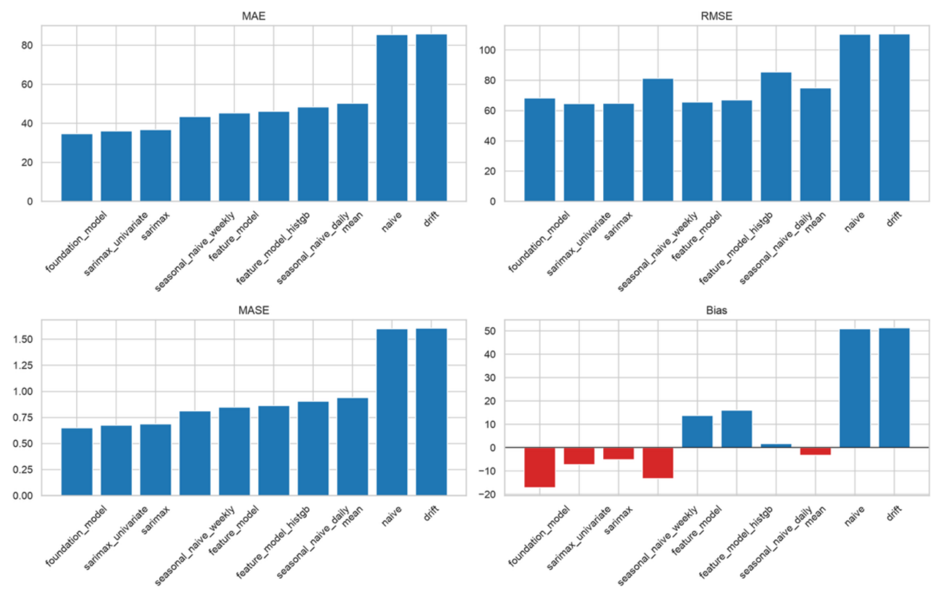


Figure 11. All four metrics, same MASE-sorted order

The above figure 11 showcases how Chronos beats all on MAE and MASE, but not on RMSE as it's higher than both SARIMAX variants. This can be contributed to RMSE penalizing large errors without proper proportions. This can also refer to its consistency in being closer to actual on a normal forecast but has a heavier tail of large misses that SARIMAX on the days it doesn't work well. Say which one is better depends on the use case whether it is accuracy or the worst-case error. Taking a close look at the bias panel adds another distinction. Chronos has the largest negative bias magnitude out of all models while the feature and seasonal naïve models tend to over forecast. This showcases how it's not just accuracy that's different, but the error character itself as well needs to be assessed as per use case.

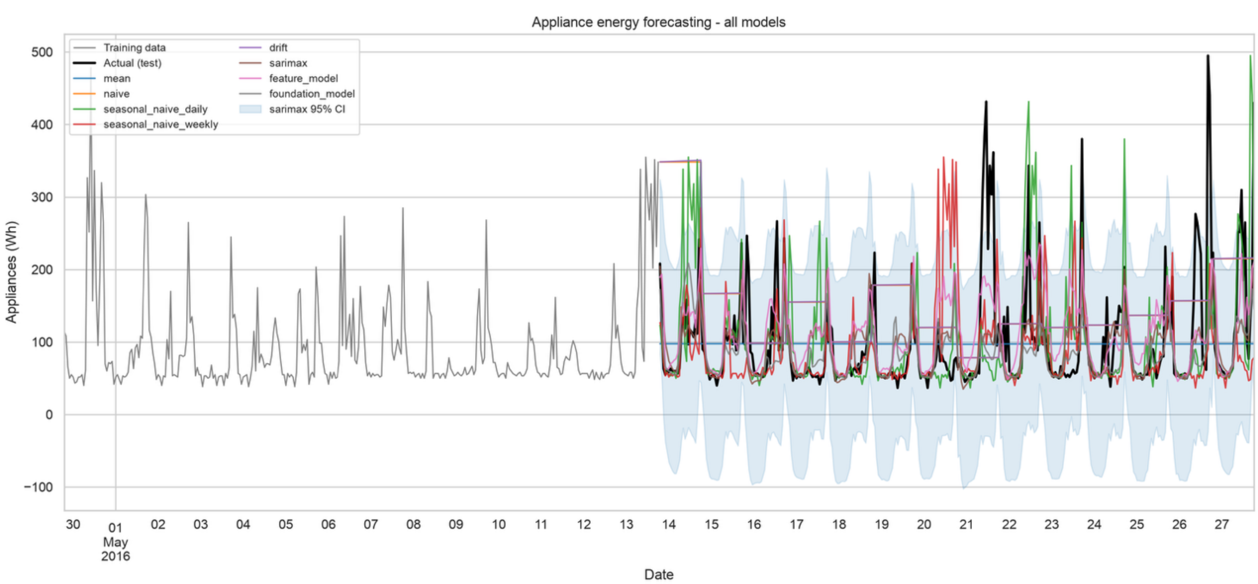


Figure 12. Forecasts from a representative subset of models against actual demand, full 14-day test period.

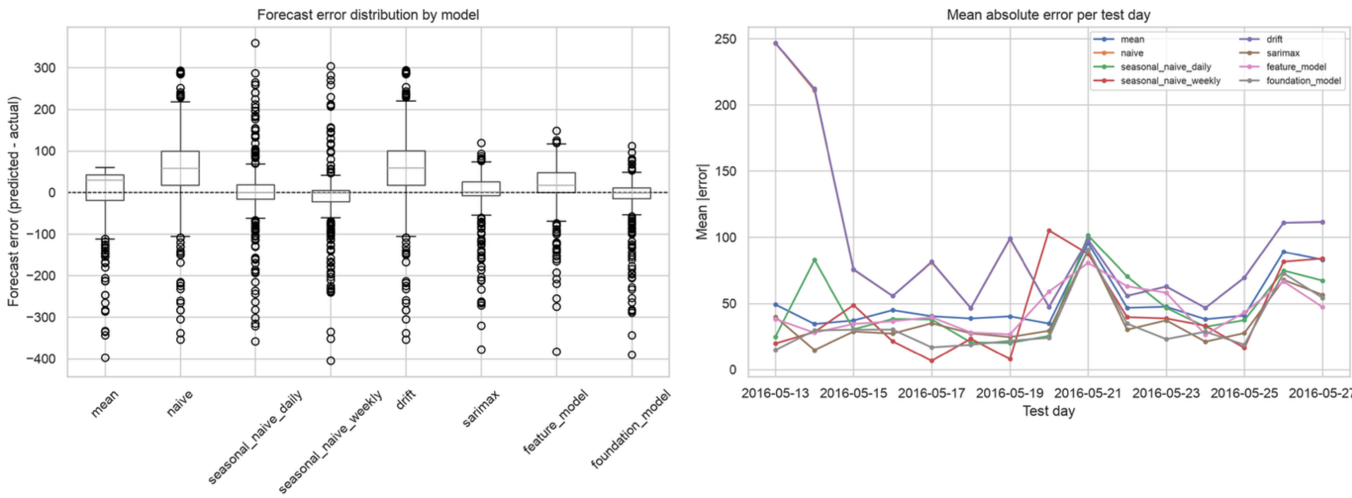


Figure 13. Per-model error distribution (predicted − actual) and daily mean absolute error across the test window.

In the above figure 13, the daily MAE panels showcase the most important key point that would help determining each model against real life scenarios. Nearly every model's error rose on 21 May. This is direct evidence on a day level that data being under forecasted is a shared systematic limitation across model families. Irregular events like power blackouts or surges usually rarely happen on daily routine times. If models are only conditioned on time of day or week, they can't anticipate irregular occurrences. This cannot be fixed and counted as an error, rather its is a structural limitation, one which can be worked on as mentioned in section 10's future work.

9. Answers to the Required Questions

1. Strongest benchmark and series structure. The weekly seasonal naïve has strongest MASE value of 0.813, while seasonal-naive follows, mean is next and then the naïve/drfit.AS the values repeat from a week ago, they outperform the yesterday's value as the whole series is a recurring daily/weekly routine and not some short-term momentum. All this is also supported by the boxplots in figure 3 where the weekday/end effect is clear. However, it is not as explicit as the hour of day effect, just a bit subtler.

2. Does SARIMAX improve on the strongest benchmark? Talking strictly on benchmark, yes. The 15-17% reduction in MASE on the already best seasonal-naive is the evidence of that. Daily seasonality and short-range correlation are much better captured than other models. Looking at Ljungs box, it is at borderline at lag 24 value and gets clean at the next 48 lag. The Q-Q plots has an upper tail deviation that shows continued under precision on large surge spikes. The exogeneous nature of the weather doesn't help and the weak covariate correlation pattern in the figure 5 explains why.

3. Does XGBoost improve as features are added, and which groups help? Adding feature does help but only if they are meaningful, just adding features for the sake of adding them can introduce unwanted signals in the data from useless feature set. Lags alone gives major improvement (0.774, less is better), when time of day is added its at (0.756). But once more features such as rolling windows without time feature are added, it hurts the overall performance (0.811). Even worser are the results when you add the sensor/weather data as the figures reach 0.849, which was the worst out of four configurations tested. All in all, lag and time features serve to be the most useful, whilst sensor/weather end up diluting the small training set, weakening the signal with redundant target correlation.

4. Does the foundation model outperform the others, and does it justify the complexity? Overall, as per the metrics mentioned in the table 3, Chronos has the best overall MASE (0.649) despite it having zero task specific training. This can be considered a substantive gain against its competition. But it still cannot be considered unconditionally better than its close competitor SARIMAX, as the RMSE is worser than both SARIMAX variants. Its advantage and real life use case lies in a typical accuracy-based case, not bounding worst case error (explained in section 8). Whether this justifies the heavier runtime is purely a deployment cost question that cannot be assume from ranking alone.

5. Which variables are genuinely known at the forecast origin? Since this is a time series problem, time features are always known in advance. Feature that come next such as lag and rolling are also available if you recursively forecast the implementation (More explained in section 6). Information such as sensor and weather values aren't known at the forecasting origin as they contribute as extra data points. The SARIMAX+exog and the full featured XGBoost both use the realized test set values which makes those two results conditional and untrue for unconditional real life scenario forecasts. From test we did notice how the covariate using variants scored worser than he covariate free counterpart anyways. The chaise of best performing model (from univariate SARIMAX or Chronos) doesn't depends on the unrealistic assumption in any way whatsoever.

6. Practical recommendation. Keeping in view all the variables, one should aim for SARIMAX as the default model for predictions for most smart home-based deployments. It captures all the achievable improvements unlike the seasonal naïve ones and also produces generally actually calibrated confidence intervals that XGBoost doesnt It also needs no external weather forecast dependency and updates can be done very cheaply via appending the data. Chronos however is still the right call in cases where maximum typical case accuracy is the main aim and having a heavier runtime is acceptable. An example of heavier runtime can be a cloud-based compute rather than edge device like phone. For those reasons, both have their own use cases. Hence there is no single winner.

10. Limitations and Future Improvements

- The data comes from a single household, and its duration is very limited (4,5 months). The data only covers the winter to spring season only with a 10-day gap in training data. There is no representative data for summer colling load. This makes it unclear whether Chronos would still beat the SARIMAX models in warmer weather or in some other household with different electrical needs. Generalization is a big issue.
- These inferences assume that the weather is known in advance which is far from realistic. If an actual system were deployed, it should docus on weather forecasts to make decisions. The XGBoost model should also only work on its best feature flag set instead of full feature set that introduces more erroneous signals. As for SARIMAX improvements, it can and should focus on grid-based search rather than fixed by hand.
- There is no way to check the native uncertainty from tree models. As seen via the results beforehand, both XGBoost and HistGB give point forecasts only. There is no interval data to compare to against Chronos or SARIMAX's confidence bands as tree based aren't implemented with that functionality. The clearest next step would be the add uncertainty using quartile regression, conformal prediction, or finetuning Chronos on this household's history so it understands the problem at hand much better as none of the models currently capture the sharp under forecasting at high usage spikes or surges.

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